

CS4037 – Introduction to Cloud Computing – Extra Credit Assignment

- *This is an optional assignment that will provide 5 extra points towards your total points.*
- *It is composed of learning modules, hand on work/automation script, and a brief exam to ensure you have to done the work yourself. This pattern is very similar to the way certifications work.*
- *You must demonstrate your work/automation script and score at least 70% to get total of 5 points.*
- *If you don't pass the written test or cannot explain your work adequately, you will not be awarded ANY points.*

Due Date: April 10th, 2025

Learning Modules

Complete the following Bicep learning modules:

- <https://learn.microsoft.com/en-us/training/paths/fundamentals-bicep>
- [Intermediate Bicep - Training | Microsoft Learn](#)
- [Advanced Bicep - Training | Microsoft Learn](#)

Complete anyone of the below, or both (if you prefer) of the following learning modules:

Option 1: [Deploy Azure resources by using Bicep and Azure Pipelines - Training | Microsoft Learn](#)

Option 2: [Deploy Azure resources by using Bicep and GitHub Actions - Training | Microsoft Learn](#)

Hands on Work

Create a modular bicep deployment template which deploys the following infrastructure:

1. 2 VNets with at least two subnets each named infra and storage.
2. Peer both VNets
3. Deploy at least one VM in each VNet's infra subnet
4. Deploy at least one ZRS storage account in each VNet's storage subnet
5. Ensure that Azure monitor is enabled for all resources
6. Deploy the bicep module using Azure Pipelines or github Actions.

Notes:

1. VNet, VM, Storage, and Azure Monitor functionalities must be in separate bicep files.
2. You can sign up for an \$100 education azure subscription to perform the above work.